Final IJRSI Audit Report

Scope

This report captures the final implementation and audit status for the IJRSI revision work in the Vivasayi AI repository. It is grounded in the repository’s existing files, preserved historical outputs, and the verified RAG behaviour in this revision.

Evidence available in repository

The repository retains the following core evidence:

? `IJRSI_REVISION_AUDIT.md` — reviewer-by-reviewer mapping to sections, evidence, missing evidence, and required follow-up.

? `RAG_ROOT_CAUSE_REPORT.md` — diagnosis of the root causes in the retrieval stack and knowledge base.

? `RAG_IMPLEMENTATION_PLAN.md` — implementation roadmap and execution order.

? `docs/research/evaluation/live/rag_retrieval_2026-10-16.md` — baseline retrieval evaluation showing poor retrieval and header-only chunk failures.

? `rag/ingestHelpers.js` — safer chunking and metadata contract, including row-bearing chunk enforcement.

? `tests/rag.chunker.unit.test.js` — deterministic expectations for row-bearing chunks and metadata quality.

? `services/chat.service.js` — runtime retrieval path.

Verified improvement implemented

We implemented a conservative retrieval safeguard in `services/chat.service.js`:

? header-only chunks are filtered out before they reach the LLM;

? chunks without row-bearing evidence are discarded;

? file attribution stays by filename only;

? API contracts and unrelated application behaviour remain intact.

This is a focused improvement aligned with the reviewer’s concern about header-only hits and non-answer-bearing chunks.

Remaining scientific constraints

The repository still has the following genuine constraints and no fabricated workaround was introduced:

? the knowledge base remains mixed between statistics, geography, and limited advisory material;

? independent agronomist validation is still absent;

? no authoritative claimant-rights service is available for real-world verification;

? the live satellite study remains a feasibility study, not a validated crop-damage detector.

These are documented limitations and are not hidden by over-claiming.

Validation state

Targeted validation was run on the retrieval and chunker path.

Result: the focused regression test file for retrieval behavior passed.

We did not claim a full benchmark improvement because the project’s wider frozen evaluation files and legacy validation scripts still contain pre-existing incompatibilities that are outside this narrow retrieval patch.

Final assessment

The revision package is appropriately conservative and evidence-based. It improves retrieval quality where the repository already showed a concrete technical defect, while explicitly preserving the original negative findings and the project’s scientific limitations.